

A rank-one completion counterexample to Conjecture 2835

Chengzhe Gao

The asserted criterion says that a rank- r matrix is uniquely determined by the entries on a graph support G if and only if G contains an r -regular supported subgraph. The sufficiency direction is false already for a 2×2 rank-one matrix.

Take

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad G = \{(1, 1), (2, 2)\}.$$

Here G is the bipartite observation graph, with two row vertices and two column vertices. Its two edges form a perfect matching, so G itself is 1-regular. In particular it contains a 1-regular subgraph whether or not “subgraph” is required to be spanning.

Theorem 1. *The entries on G do not uniquely recover A among rank-one matrices.*

Proof. Both matrices have determinant zero and a nonzero entry, hence rank exactly one. More explicitly, $A = (1, 1)^T(1, 1)$ and $B = (1, -1)^T(1, -1)$. Their two observed diagonal entries agree. Their $(1, 2)$ entries differ, so they are distinct completions of the same observations, each of the specified rank. Since G itself is 1-regular, the conjectured sufficient condition holds while unique recovery fails. $\square$

The entries are rational, and the same matrices have rank one over the real and complex fields. The example does not impose an unintended spanning condition on the source. It also does not use a counting heuristic as a substitute for constructing two completions. Failure of this necessary part of the conjecture suffices, independently of the later observation-count claim.

Formal verification. `Matrix` is the full space of 2×2 matrices over `Std.Internal.Rat`. The function `rank` is the determinantal rank: two if the sole full-size determinant is nonzero, zero if every one-by-one minor is zero, and one otherwise. This is exactly the standard rank definition by the largest nonvanishing minor in dimension two. The proof also checks every selected two-by-two minor and an explicit nonzero one-by-one minor for each matrix, as well as both outer-product descriptions.

The support is an actual Boolean bipartite graph. Row and column degrees count all neighbors, and the regular-subgraph predicate permits arbitrary selected nonempty row and column vertex sets, with edges restricted to those sets and to G . The witness selects every vertex and G itself. Unique recoverability quantifies over *all* rational matrices of the same rank, not a finite candidate table. The program constructs B as a counterexample to that universal quantifier. The final theorem negates the rank-one, two-by-two instance of the source’s iff criterion. Lean 4.19.0 checks this with `Std`, no unproved declarations, and no native evaluation tactic.